EUROGRIP × UNSTOP COLLABORATION

Case Study 2

Winning the EV Two-Wheeler Opportunity Through Next-Gen Tyre Technology

Background

Electric two-wheelers place fundamentally different demands on tyres than internal-combustion bikes. Instant motor torque changes acceleration-phase wear patterns; regenerative braking creates different heat and stress cycles; silent motors make tyre noise (NVH) far more noticeable to riders; and battery weight increases load requirements. Meanwhile, rolling resistance directly affects vehicle range — making the tyre a lever on the EV's core value proposition, not just a wear component.

As EV adoption accelerates across two-wheeler categories, there is a clear opportunity for a tyre brand to lead with technology purpose-built for this shift, rather than adapting existing ICE-era tyre designs.

The Challenge

How should Eurogrip design a next-generation tyre concept purpose-built for EV two-wheelers — addressing compound composition, tread design, and load rating — that meets these engineering demands while remaining manufacturable and commercially viable at scale?

Technical Areas to Address

- Tyre construct: balancing low rolling resistance (for range) against grip and durability under instant-torque acceleration.
- Tread pattern: optimising for reduced road noise (NVH) and even wear under regenerative-braking load cycles.
- Optimized performance on vital needs like grip, high speed stability, damage resistance, durability.
- Load rating: accommodating higher kerb weight from battery packs without compromising handling.
- ~~Thermal management: addressing heat build-up from frequent regenerative-braking cycles.~~

Constraints

Global viability	Must be a globally viable product, meeting internationally accepted tyre compliance and testing standards — not India-only — since Eurogrip sells across multiple markets.
Cost realism	Target cost-per-tyre must stay within a realistic commercial band for the mass-market EV segment, not a premium/niche positioning.
Manufacturability	Concept must be manufacturable on largely existing tyre-plant processes, not requiring a wholly new production line.

Format & Timeline

Round 1 — Executive Summary + Video Submission

- 1-page executive summary of the technical concept and direction.

- 2-minute video pitch — precise and direction-focused, not a polished production. The purpose is to convey clarity of thinking, not creativity or presentation quality at this stage.

Round 2 — Detailed Submission (1–2 weeks)

- A technical deck of up to 15 slides (excluding annexures), covering: the engineering concept, compound and tread rationale, compliance approach, and cost/manufacturability feasibility.

Evaluation Criteria

These criteria apply to the Round 2 submission and are meant to help judges score entries consistently.

Criterion	What we're looking for
Technical soundness	Is the engineering approach sound and well-reasoned — grounded in actual material science and tyre-design principles?
EV-specific understanding	Does the concept demonstrate genuine understanding of how EV two-wheeler demands differ from ICE (torque, regen braking, NVH, load)?
Innovation	Is there a clear point of differentiation from existing tyre technology, not just a minor tweak to current designs?
Manufacturability & cost realism	Could this concept realistically be produced at the target cost, using largely existing processes?
Clarity of communication	Is the technical thesis communicated clearly and precisely, especially in the Round 1 video?